

CHEATSHEET · QUICK REFERENCE

# Lab 30 — Storage Troubleshooting

Diagnose disk-full vs. inode-full, recover a corrupt superblock, rebuild a degraded RAID, and read cache-battery, cable and swap symptoms.

## Capacity and I/O

```
df -h ; df -i
Bytes vs. inodes (full inodes != full bytes)

iostat -xz 1 5
High %util + await = saturated disk

iotop -bo -n 3
Who is hammering the disk

mount | grep -E 'noatime|relatime'
atime overhead check
```

## Filesystem recovery

```
mkfs.ext4 -n $L00P
Print backup superblock locations

fsck.ext4 -b 32768 $L00P
Repair using a backup superblock

sgdisk -e /dev/sdX
Rebuild GPT from the backup header

testdisk
Rebuild a corrupt MBR
```

## RAID failure and rebuild

```
cat /proc/mdstat
Array state

mdadm /dev/md30 --fail /dev/loop32 --remove /dev/loop32
Fail and remove a drive

mdadm /dev/md30 --add /dev/loop35
Add a spare (rebuild)
```

## Boot / cache / swap symptoms

| Symptom                      | Cause / fix                            |
|------------------------------|----------------------------------------|
| OS not found                 | BIOS order / dead disk                 |
| GRUB rescue                  | /boot gone -> rescue + reinstall GRUB  |
| Sudden write slowdown        | Cache battery/BBU bad -> write-through |
| Repeated hard reset in dmesg | Bad cable / backplane -> reseal        |
| Thrash / OOM                 | Enlarge swap or add RAM                |